# Problem Set 3

## Section 3.2

18) $x - 3y - z = 12$

$x = 12 + 3y + z$

Therefore $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

24) Such a matrix does not exist. Suppose such a matrix did exist, lets name it A. Then A must be a 3x3 matrix. Also, since the nullspace has two independent vectors, there are at least two free colums and hence 1 or 0 pivot columns. But the column space has two independent vectors hence at least 2 pivot columns i.e. a contradiction and so our supposition is false.

36) $N(C) = N(A) \cap N(B)$ since $Cx = \begin{bmatrix} Ax \\ Bx \end{bmatrix}$ and so when $Cx = 0$, ~~Ax~~ we ~~also~~ also have $Ax = 0$ and $Bx = 0$

~~Problem 3.5e~~

37)

Node 1:- $y_1 + y_4 = y_3$

$y_1 + 0y_2 - y_3 + y_4 + 0y_5 + 0y_6 = 0$

Node 2:- $y_5 + y_2 = y_1$

$-y_1 + y_2 + 0y_3 + 0y_4 + y_5 + 0y_6 = 0$

Node 3:- $y_3 + y_6 = y_2$

$0y_1 - y_2 + y_3 + 0y_4 + 0y_5 + y_6 = 0$

Node 4:- $0y_1 + 0y_2 + 0y_3 - y_4 - y_5 - y_6 = 0$

In $Ay = y$ form:-

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Perform rref of A:-

$$\begin{bmatrix} 1 & 0 & -1 & 0 & -1 & -1 \\ 0 & 1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$N = \begin{bmatrix} -F \\ I \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & -1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(if columns 3 and 4 switched places of A)

To adjust for the fact that this N is for A with colums 3 and 4 switched, we switch rows 3 and 4 of N:-

$$N = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The columns of this matrix are our special solutions.

## Section 3.3

19) Suppose A and B are $n \times n$ matrices such that $AB = I$.

Given $\text{rank}(AB) \leq \text{rank}(A)$, we know that $n \leq \text{rank}(A)$ (Since AB is the $n \times n$ I matrix and so has rank n).

Now $n < \text{rank}(A)$ is not possible because A is $n \times n$ and so cannot have rank greater than n, therefore $n = \text{rank}(A)$.

Since A is square and full rank, A is invertible, and

25) $A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 1 & 2 & 2 & 5 \\ 1 & 3 & 2 & 6 \end{bmatrix}$

rref of $A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Hence $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

27)

a) I is $r \times r$ F is $r \times (n-r)$
Bottom left 0 is $(m-r) \times r$
Bottom right 0 is $(m-r) \times (n-r)$

b) Right inverse:- $\begin{bmatrix} I_{r \times r} \\ 0_{(n-r) \times r} \end{bmatrix}$

c) Left inverse = $[\, I_{r \times r} \; 0_{r \times (m-r)} \,]$

d) $R^T = \begin{bmatrix} I_{r \times r} & 0 \\ F^T_{(n-r) \times r} & 0 \end{bmatrix}$ → $r \times (m-r)$, → $(n-r) \times (m-r)$

Since F is in the span of I, the rref is:- $\begin{bmatrix} I_{r \times r} & 0_{r \times (m-r)} \\ 0_{(n-r) \times r} & 0_{(n-r) \times (m-r)} \end{bmatrix}$

e) $R^TR = \begin{bmatrix} I_{r\times r} & F_{r\times(n-r)} \\ F^T_{(n-r)\times r} & F^TF_{(n-r)\times(n-r)} \end{bmatrix}$

Since row 2 = $F^T$(row 1), rref is:-

$$\begin{bmatrix} I_{r\times r} & F_{r\times(n-r)} \\ O_{(n-r)\times r} & D_{(n-r)\times(n-r)} \end{bmatrix}$$

Suppose R is in rref and we have shown in part e that the rref of $R^TR$ is R. Since row operations do not change the nullspace, it follows that $N(R^TR) = N(R)$.

28) Since columns of F are linearly dependent on colums of I, we can eleminate F completely and hence get:-

$$\begin{bmatrix} I_{r\times r} & O_{r\times(n-r)} \\ O_{(m-r)\times r} & O_{(m-r)\times(n-r)} \end{bmatrix}$$

17) Suppose column j is a column of a matrix B such that column j is a combination of previous columns of B:-

$b_j = c_1b_1 + c_2b_2 + \cdots + c_{j-1}b_{j-1}$

where $c_i$ is a scalar constant

By definition of matrix multiplication, column $j$ of $AB$ (i.e. $ab_j$) is $Ab_j$ (i.e. matrix $A$ multiplying $b_j$) and since $b_j = c_1b_1 + c_2b_2 + \cdots + c_{j-1}b_{j-1}$ we have $Ab_j = A(c_1b_1 + c_2b_2 + \cdots + c_{j-1}b_{j-1})$

$$= c_1Ab_1 + c_2Ab_2 + \cdots + c_{j-1}Ab_{j-1}$$

And since $Ab_i = ab_i$ (i.e. column $i$ of matrix $AB$), we get:-

$$ab_j = c_1ab_i + c_2ab_2 + \cdots + c_{j-1}ab_{j-1}$$

b) Let $A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $A_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

$A_1B = B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ which has rank 1

$A_2B = A_2$ which has rank 0.

## Section 3.4

13) a) This is false because $x_p$ may not be a particular solution anymore if it is scaled by a scalar that is not 1.

b) This is only true if $A$ has full column rank, otherwise $x_n$ has values other than the zero vector and each such $x_n$ gives a different answer for $x_p + x_n$.

c) Let $A = \begin{bmatrix} 2 & -10 \\ -1 & 5 \end{bmatrix}$ and let $b = \begin{bmatrix} 50 \\ -25 \end{bmatrix}$.

With all free variables set to 0, the solution is $\begin{bmatrix} 25 \\ 0 \end{bmatrix}$ which has length 25, whereas another solution is $\begin{bmatrix} 0 \\ -5 \end{bmatrix}$ which has length 5.

d) The origin is still in the nullspace

25)

a) $r < m$

b) $r = m \quad r < n$

c) $r = n \quad r < m$

d) $r = m = n$

28) $[U\ 0] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 0 & 4 & 0 \end{array}\right]$

$\downarrow$

$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$

$\downarrow$

rref :- $\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$

Solving for $x_2 = 1$ we get:-

$x_1 + 2(1) + 0 = 0 \Rightarrow x_1 = -2$

$x_3 = 0 \Rightarrow x_3 = 0$

$x_n = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ and scalar multiples of it describe $\mathcal{C}$ is any scalar multiple.

$[U\ c] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 0 & 4 & 8 \end{array}\right]$

$\downarrow$

$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right]$

$\downarrow$

Rref:- $\left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$

Solving for $x_2 = 0$ we get:-

$x_1 + 2(0) + 0\,x_3 = -1 \Rightarrow x_1 = -1$

$x_3 = 2 \Rightarrow x_3 = 2$

$x_p = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$

36) Yes it must be true. Let $y$ be any vector ~~such that~~ and let $b$ be any vector such that $Ay = b$. Then by our given supposition $Cy = b$ and so $Ay = Cy$ and so $A = C$.